AI/ML Interview Task: Intelligent OCR System Using Open-Source LLM

Objective

Develop an Intelligent OCR (Optical Character Recognition) System using an open-source Large Language Model (LLM) that can extract, understand, validate, and structure information from document images or PDFs.

Note: The use of proprietary APIs such as OpenAI, Gemini, Claude, or any paid OCR services is not allowed. The solution must be developed using open-source OCR and LLM models.

Problem Statement

Build an AI application that accepts document images or PDFs and performs intelligent OCR with document understanding. The system should not only extract text but also analyze the document using an LLM to generate structured information and answer user queries.

Functional Requirements

1. Image Preprocessing

Implement preprocessing techniques to improve OCR accuracy:

Grayscale conversion, Noise removal, Thresholding, Deskewing, Image enhancement (optional)

2. OCR Extraction

Use any open-source OCR library such as:

EasyOCR, PaddleOCR, Tesseract OCR, TrOCR

Extract all readable text from the document.

3. Intelligent Information Extraction Using an Open-Source LLM

Use an open-source LLM (Llama, Mistral, Gemma, Phi, Qwen, etc.) to:

Correct OCR errors.

Understand document context.

Extract important fields.

Convert extracted information into structured JSON.

Example fields:

Document Title, Name, Date, Invoice Number, Total Amount, Address, Phone Number, Email

Any other relevant information based on the document type

4. Document Classification

Automatically classify the uploaded document into categories such as:

Invoice, Receipt, Resume, Aadhaar Card, PAN Card, Driving License, Passport, Bank Statement, Other

Display the confidence score (if available).

5. Data Validation

Validate extracted information using suitable rules.

Examples:

Full Name, Email format, Phone number, Date format, PAN format, GST Number, Aadhaar Number (format validation only), Numeric values

7. Structured Output

Original Image

Extracted OCR Text

Corrected Text

Document Type

Extracted JSON

Validation Results

User Question & AI Response

Technical Requirements

Programming Language: Python **Micro Integrated Semiconductor Systems Pvt. Ltd.**

Frontend: Streamlit or Gradio

OCR: EasyOCR / PaddleOCR / Tesseract / TrOCR

LLM: Open-source models from Hugging Face (Llama, Gemma, Phi, Mistral, Qwen, etc.)

Image Processing: OpenCV

Framework: PyTorch or Transformers

Bonus Features (Optional)

Multi-page PDF support

Handwritten text recognition

Table extraction

Bounding box visualization

Confidence score visualization

Export extracted data to JSON or CSV

Docker deployment

Deliverables

Submit the following:

- GitHub Repository
- Source Code
- README with installation and execution steps
- Sample input documents
- Output screenshots
- Architecture diagram
- Short technical report (2–5 pages) explaining:
- OCR workflow
- LLM integration
- Information extraction approach
- Validation logic
- Challenges faced

Important Instructions

- Do not use proprietary AI APIs such as OpenAI, Gemini, Claude, or paid OCR services.
- Use only open-source OCR libraries and open-source LLMs.
- Clearly document your implementation, assumptions, and limitations.
- Write clean, modular, and well-documented code.

This assignment is designed to evaluate the candidate's practical knowledge in:

- Computer Vision and Image Processing
- OCR Techniques
- Natural Language Processing (NLP)
- Open-source Large Language Models (LLMs)
- Prompt Engineering
- Python Development
- Information Extraction
- AI Workflow Design
- Building an end-to-end intelligent document processing application suitable for real-world use.

Share Task on: Email id : sadhegaonkar@sevenmentor.com